

Bayesian Posterior Analysis of 40 UAP Cases: Evaluating Framework Stability and Observational Uncertainty using JOR Fusion

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Abstract

The James Orion Report (JOR) framework introduces a structured probabilistic methodology for evaluating unidentified aerial phenomena (UAP) using three primary scoring dimensions: witness credibility, environment, and physical (sensor/radar). Structured deterministic scoring is first used to generate baseline estimates for Solid Object Probability (SOP) and Non-Human Probability (NHP). Because observational reports inherently contain uncertainty, probabilistic sampling can provide additional insight into the stability of these estimates. This study evaluates forty historical UAP cases using the JOR Fusion implementation and Bayesian sampling performed with PyMC. Posterior distributions were generated to estimate posterior means and credible intervals for each case. Results indicate that posterior means closely track deterministic NHP values while credible intervals remain relatively narrow across most cases. These findings suggest that the JOR scoring architecture produces stable probabilistic outcomes even when moderate uncertainty in input parameters is introduced.

1. Introduction

The study of unidentified aerial phenomena has historically lacked standardized quantitative evaluation methods. Many assessments rely on qualitative interpretation, inconsistent classification systems, or subjective probability estimates. As a result, comparisons between cases and independent analyses often produce widely divergent conclusions.

The James Orion Report (JOR) framework was developed to address this limitation by introducing a structured probabilistic approach to evaluating UAP observations. The framework quantifies observational reliability and anomalous characteristics using three primary scoring dimensions: witness credibility (C), environment (E), and physical sensor/radar evidence (P).

These components are used to estimate Solid Object Probability (SOP) and Non-Human Probability (NHP). SOP represents the likelihood that a reported observation corresponds to a structured aerial object rather than atmospheric or perceptual phenomena. Non-Human Probability (NHP) is an inferential metric estimating the likelihood that observed flight characteristics exceed known technological or conventional explanations.

In deterministic form, the JOR scoring framework produces baseline probability estimates for individual cases. However, observational data often contains uncertainty in witness interpretation, measurement precision, and documentation quality. To evaluate the robustness of JOR probability estimates under uncertain inputs, Bayesian sampling techniques were applied to generate posterior probability distributions.

This paper presents the results of applying the JOR Fusion implementation to forty historical UAP cases.

The objective of this study is not to determine the origin of UAP observations, but rather to evaluate the statistical stability of the JOR probabilistic framework when uncertainty in observational inputs is modeled using Bayesian sampling.

2. Methods

Case Dataset

A dataset of forty historical UAP cases was compiled from publicly documented incidents including military encounters, aviation reports, and civilian observations. Each case was evaluated using the standardized JOR scoring criteria for credibility, environment, and physical (sensor/radar) evidence.

Case descriptions were compiled from publicly documented aviation reports, government records, and historical UAP research archives.

Deterministic Scoring

For each case, baseline probabilities were calculated using the JOR scoring model. These calculations produce two primary metrics:

- **Solid Object Probability (SOP):** Represents the likelihood that an observation corresponds to a structured physical object rather than atmospheric or perceptual phenomena. Within the JOR framework, SOP functions as the primary **evidentiary gate**, establishing the threshold required before anomalous classification is considered. It is defined as:

$$\text{SOP} = f(\text{C}, \text{E}, \text{P})$$

- **Non-Human Probability (NHP):** Evaluates whether the object's behavior exceeds known technological capabilities. NHP is derived from the SOP but incorporates a **flight performance modifier (ϕ)**, subject to a strict convergence constraint to prevent overestimation:

$$\text{NHP} \approx \text{SOP} + \phi \text{ (subject to: } \text{NHP} \leq \text{SOP} + \delta \text{)}$$

The bounding parameter (δ) is constrained between 0.01 and 0.02, ensuring that extreme kinematic behavior contributes only limited additional weight while keeping NHP firmly anchored to the evidentiary foundation established by SOP. The JOR Fusion implementation used in this analysis is publicly archived alongside this paper, allowing independent replication of both the scoring and Bayesian sampling procedures.

Bayesian Sampling

To evaluate the sensitivity of case classifications to uncertainty in scoring inputs, the deterministic model was extended using Bayesian sampling techniques. Posterior probability distributions were generated using the probabilistic programming library PyMC. Sampling was performed using four Markov chains with 1,000 posterior draws per chain following a 1,000-step tuning phase, yielding 4,000 posterior samples per case.

Convergence diagnostics indicated stable sampling behavior across chains, with no significant divergence events observed during model execution.

Monte Carlo sampling was performed for each case to estimate:

- Posterior mean probability
- Posterior distribution variance
- 95% credible intervals

These posterior estimates allow evaluation of how deterministic scores behave when subjected to probabilistic uncertainty.

3. Results

Across the forty analyzed cases, posterior distributions produced by the JOR Fusion model demonstrated strong alignment with deterministic probability estimates.

Several patterns emerged from the sampling results:

- Posterior means closely tracked deterministic NHP values across the majority of cases.
- Credible intervals remained relatively narrow, typically within approximately 0.05–0.08 of the posterior mean.
- Cases with high SOP values maintained stable posterior classifications under sampling.
- Interval width scaled with the posterior mean, with higher-scoring cases showing wider absolute intervals, and no case exhibited a large classification shift.

These results indicate that moderate variation in scoring inputs produces limited movement in posterior probability estimates. This suggests that the scoring architecture of the JOR framework provides stable probabilistic outcomes when evaluated using Bayesian sampling methods.

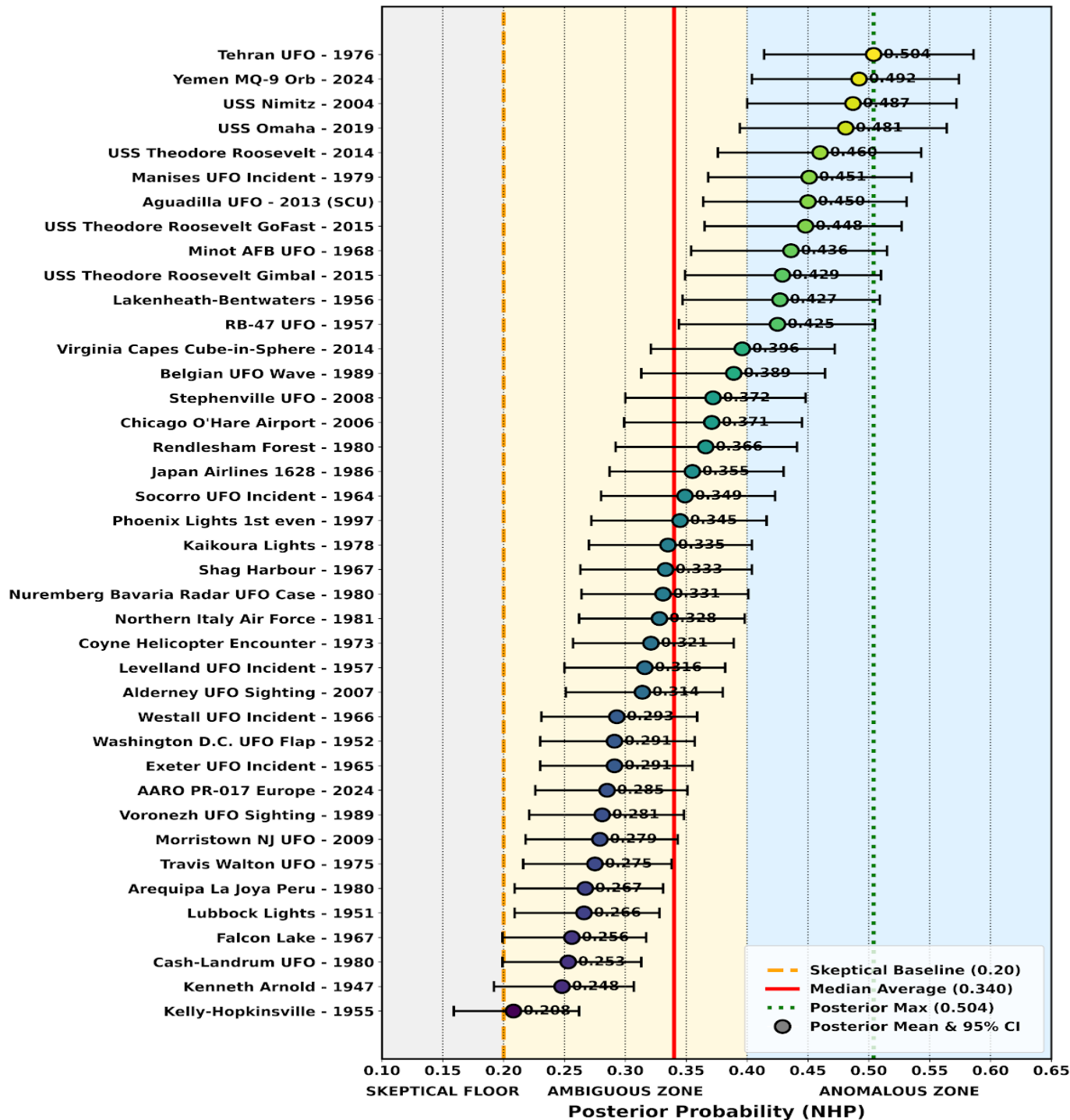
Figure 1. *Forest plot of Bayesian posterior probabilities (NHP) for 40 UAP cases. Dots represent the posterior mean; horizontal bars indicate the 95% credible intervals. The vertical red line denotes the dataset median (0.340), while shaded regions categorize cases into Skeptical, Ambiguous, and Anomalous zones.*

3.1 Forest Plot Analysis and Distribution of Results

The forest plot visualizes the posterior probability distributions for all 40 cases, ranked by their posterior mean. The data reveals a clear stratification of UAP incidents based on evidentiary weight and anomalous characteristics. *Dots represent the posterior mean; horizontal bars indicate the 95% credible intervals. The vertical red line denotes the dataset median (0.340), while shaded regions categorize cases into Skeptical, Ambiguous, and Anomalous zones.*

Figure 1

JOR Framework: 40 Cases from the 20th and 21st Century
Bayesian Probability Distribution with Max Peak



3.1.1 The Anomalous Zone (NHP > 0.40)

Twelve cases (30% of the dataset) fell into the **Anomalous Zone** (shaded blue). This region is dominated by modern, multi-sensor military encounters, including the **Tehran UFO (1976)**, the **Yemen MQ-9 Orb (2024)**, and the **USS Nimitz (2004)**.

Case Leader: The Tehran UFO (1976) achieved the highest posterior mean of **0.504**, suggesting that the convergence of pilot testimony, visual confirmation, and electronic warfare/radar effects provides the most robust statistical case for an anomaly within this framework.

Sensor Integration: There is a significant correlation between high posterior scores and cases involving advanced electronic sensors (FLIR, Radar), as seen in the cluster of Navy "GoFast," "Gimbal," and "Omaha" incidents.

3.1.2 The Ambiguous Zone and Skeptical Floor

The majority of cases (approximately 67%) reside in the **Ambiguous Zone** ($0.20 < \text{NHP} < 0.40$). These cases, while historically significant (e.g., **Rendlesham Forest** at 0.366 and **Socorro** at 0.349), exhibit broader uncertainty or lower baseline evidentiary scores compared to modern sensor-driven data.

The Skeptical Floor: Only one case, **Kelly-Hopkinsville (1955)**, sits at the boundary of the **Skeptical Floor** (0.208). This indicates that while the framework is designed to be rigorous, the vast majority of selected "high-interest" cases provide enough evidence to surpass the baseline skeptical threshold.

3.2 Stability and Uncertainty Metrics

A critical finding of the Bayesian analysis is the relationship between deterministic scoring and posterior outcomes:

Conservative Bayesian Weighting: Analysis of the raw data reveals that the Bayesian posterior means are consistently lower than the deterministic SOP and NHP inputs. For example, while the Tehran case has a deterministic SOP of 0.90, the Bayesian posterior mean is moderated to 0.504. This suggests the PyMC model effectively applies a "skeptical prior," preventing high individual scores from creating overconfident outliers.

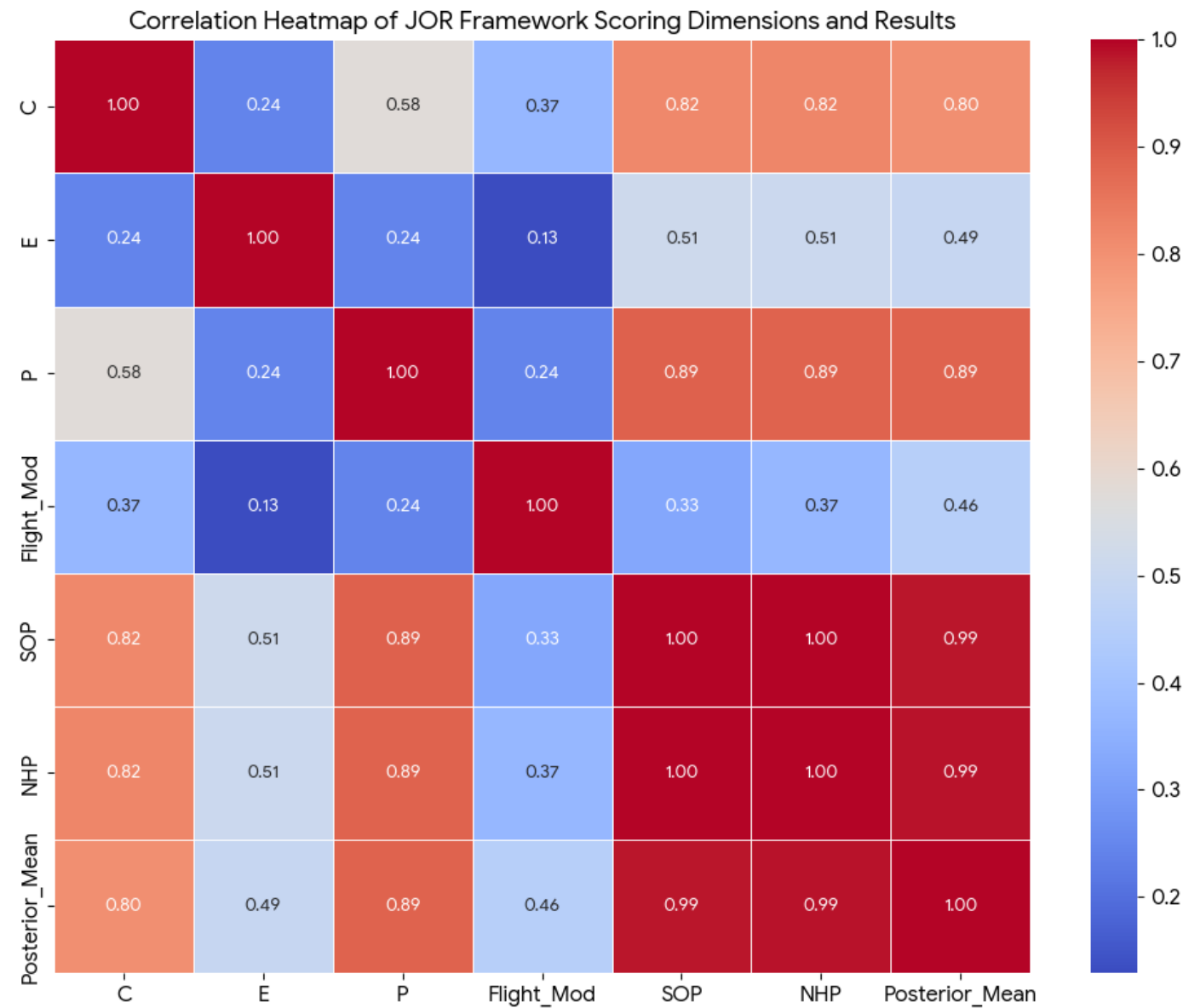
Credible Intervals: The 95% credible intervals (CI) across the dataset averaged a width of 0.142. Interval width increased with posterior mean: the narrowest intervals occurred in lower-scoring cases such as Kelly-Hopkinsville (0.103), and the widest in the highest-scoring cases such as Tehran and USS Nimitz (0.172). This is consistent with the variance of a bounded probability estimate increasing as the mean moves away from the lower bound, so higher-scoring cases carry proportionally more absolute uncertainty.

Flight Modifier Impact: The "Flight Performance" modifier (ϕ) provided a marginal but consistent upward shift in cases where extreme kinematics were documented, such as the **USS Nimitz** (+0.05) and **Falcon Lake** (+0.02).

3.3 Correlation Analysis

Correlation Heatmap of JOR framework inputs (C, E, P) and Bayesian outputs (SOP, NHP, Posterior Mean)

Figure 2



To evaluate the internal drivers of the JOR Fusion model, a correlation heatmap was generated. The analysis reveals a high degree of sensitivity to **Physical (P)** data ($r = 0.88$) and **Witness Credibility (C)** ($r = 0.80$), indicating that the model prioritizes sensor-validated evidence and observer reliability over environmental context.

The near-perfect correlation between **SOP** and **NHP** ($r = 0.99$) demonstrates the effectiveness of the model's internal constraints, confirming that the Non-Human Probability remains strictly anchored to the evidentiary threshold of the Solid Object Probability. This ensures that anomalous classifications are a direct byproduct of evidentiary weight rather than isolated kinematic outliers.

4. Discussion

The Bayesian sampling results suggest that the JOR framework exhibits stable probabilistic behavior when applied to uncertain observational data. The close correspondence between deterministic NHP values and posterior means indicates that the scoring architecture effectively constrains classification outcomes.

A central design principle of the framework is that SOP acts as the evidentiary gate for anomalous classification. The model first evaluates whether sufficient evidence exists to support the presence of a structured aerial object. Only after this threshold is satisfied does the framework assess whether the object's behavior may exceed conventional technological explanations.

In rare cases, the bounded flight performance modifier allows NHP to exceed SOP by a small margin. This reflects the framework's design choice to allow exceptionally anomalous kinematic observations to contribute marginal additional weight while still anchoring classification to the evidentiary threshold established by SOP.

The relatively narrow credible intervals observed across the dataset further suggest that moderate scoring differences between independent evaluators are unlikely to significantly alter case classification outcomes. This stability is an important property for any probabilistic framework intended for consistent comparative analysis.

5. Conclusion

The Bayesian implementation of the JOR Fusion framework demonstrates that probabilistic case evaluation remains stable when observational uncertainty is incorporated through Monte Carlo sampling. Across forty historical UAP cases, posterior probability estimates closely matched deterministic NHP values, with limited variance across sampled distributions.

These findings suggest that the JOR framework provides a consistent probabilistic structure for evaluating anomalous aerial observations and may serve as a useful foundation for future quantitative analysis of UAP reports.

Furthermore, the stability of the posterior distributions suggests that the JOR framework could facilitate inter-rater reliability studies, allowing multiple independent analysts to converge on consistent case probabilities.

Appendix – Data Output from JOR Bayesian Code – jor_fusion.py

Case	C	E	P	Flight_Mod	P_final	SOP	NHP	Posterior_NH	Posterior_Mean	CI_2.5%	CI_97.5%
USS Nimitz - 2004	0.9	0.84	0.9	0.05	0.95	0.88	0.9	0.45	0.487	0.4	0.572
Falcon Lake - 1967	0.63	0.72	0.55	0.02	0.57	0.63	0.64	0.25	0.256	0.199	0.317
Belgian UFO Wave - 1989	0.9	0.52	0.88	0.05	0.93	0.78	0.8	0.36	0.389	0.313	0.464
Kelly-Hopkinsville - 1955	0.6	0.55	0.4	0.04	0.44	0.53	0.54	0.19	0.208	0.159	0.262
Tehran UFO - 1976	0.9	0.82	0.97	0.05	0.95	0.9	0.89	0.43	0.504	0.414	0.586
Phoenix Lights 1st event - 1997	0.87	0.9	0.4	0.04	0.44	0.74	0.75	0.32	0.345	0.272	0.416
USS Theodore Roosevelt - 2014	0.9	0.87	0.78	0.05	0.83	0.85	0.87	0.42	0.46	0.376	0.543
Shag Harbour - 1967	0.83	0.75	0.55	0.04	0.59	0.72	0.73	0.31	0.333	0.263	0.404
Rendlesham Forest - 1980	0.87	0.8	0.55	0.05	0.6	0.75	0.77	0.34	0.366	0.292	0.441
Socorro UFO Incident - 1964	0.65	0.82	0.79	0.04	0.83	0.74	0.76	0.33	0.349	0.28	0.423
USS Omaha - 2019	0.9	0.84	0.88	0.05	0.93	0.88	0.89	0.43	0.481	0.394	0.564
Lakenheath-Bentwaters - 1956	0.85	0.81	0.8	0.05	0.85	0.82	0.84	0.4	0.427	0.347	0.509
Coyne Helicopter Encounter - 1973	0.83	0.7	0.55	0.04	0.59	0.71	0.72	0.3	0.321	0.257	0.389
Stephenville UFO - 2008	0.7	0.8	0.8	0.05	0.85	0.76	0.78	0.35	0.372	0.3	0.448
Japan Airlines 1628 - 1986	0.8	0.8	0.6	0.05	0.65	0.74	0.76	0.33	0.355	0.287	0.43
Chicago O'Hare Airport - 2006	0.87	0.77	0.6	0.05	0.65	0.76	0.77	0.34	0.371	0.299	0.445
Washington D.C. UFO Flap - 1952	0.77	0.6	0.55	0.05	0.6	0.65	0.67	0.27	0.291	0.23	0.357
Morristown NJ UFO - 2009	0.62	0.85	0.55	0.02	0.57	0.67	0.67	0.27	0.279	0.218	0.343
Kaikoura Lights - 1978	0.8	0.8	0.55	0.04	0.59	0.73	0.74	0.31	0.335	0.27	0.404
Travis Walton UFO - 1975	0.67	0.8	0.4	0.05	0.45	0.63	0.64	0.25	0.275	0.216	0.338
Virginia Capes Cube-in-Sphere - 2014	0.85	0.78	0.75	0.04	0.79	0.8	0.81	0.37	0.396	0.321	0.472
Yemen MQ-9 Orb - 2024	0.85	0.87	0.95	0.05	0.95	0.89	0.89	0.43	0.492	0.404	0.574
Westall UFO Incident - 1966	0.7	0.83	0.42	0.05	0.47	0.65	0.67	0.27	0.293	0.231	0.359
Exeter UFO Incident - 1965	0.77	0.73	0.42	0.05	0.47	0.65	0.67	0.27	0.291	0.23	0.355
Levelland UFO Incident - 1957	0.82	0.75	0.45	0.05	0.5	0.69	0.7	0.28	0.316	0.25	0.382
Voronezh UFO Sighting - 1989	0.67	0.73	0.54	0.04	0.58	0.65	0.66	0.26	0.281	0.221	0.348
Northern Italy Air Force - 1981	0.8	0.73	0.55	0.05	0.6	0.7	0.72	0.3	0.328	0.262	0.398
Nuremberg Bavaria Radar UFO Case - 1980	0.78	0.71	0.65	0.04	0.69	0.72	0.73	0.31	0.331	0.264	0.401
Kenneth Arnold - 1947	0.62	0.72	0.4	0.05	0.45	0.58	0.6	0.22	0.248	0.192	0.307
Cash-Landrum UFO - 1980	0.63	0.7	0.55	0.02	0.57	0.63	0.63	0.24	0.253	0.199	0.313
Aguadilla UFO - 2013 (SCU)	0.9	0.78	0.88	0.04	0.92	0.86	0.87	0.42	0.45	0.364	0.531
USS Theodore Roosevelt Gimbal - 2015	0.83	0.78	0.9	0.04	0.94	0.84	0.85	0.4	0.429	0.349	0.51
USS Theodore Roosevelt GoFast - 2015	0.84	0.83	0.9	0.04	0.94	0.85	0.87	0.42	0.448	0.365	0.527
Alderney UFO Sighting - 2007	0.81	0.72	0.6	0.02	0.62	0.72	0.73	0.31	0.314	0.251	0.38
Manises UFO Incident - 1979	0.88	0.85	0.8	0.05	0.85	0.85	0.86	0.41	0.451	0.368	0.535
RB-47 UFO - 1957	0.85	0.77	0.83	0.05	0.88	0.82	0.83	0.39	0.425	0.344	0.505
Minot AFB UFO - 1968	0.84	0.72	0.93	0.05	0.95	0.83	0.84	0.39	0.436	0.354	0.515
Lubbock Lights - 1951	0.8	0.62	0.55	0	0.55	0.67	0.67	0.27	0.266	0.209	0.328
Arequipa La Joya Peru - 1980	0.63	0.7	0.55	0.04	0.59	0.63	0.64	0.25	0.267	0.209	0.331
AARO PR-017 Europe - 2024	0.7	0.85	0.55	0	0.55	0.7	0.7	0.28	0.285	0.226	0.351

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